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MILITARY SPECIFICATION

SWITCHES, LIQUID LEVEL, GENERAL SPECIFICATION FOR

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design requirements and all performance requirements for procurement of switches for liquid level control and warning systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-S-3136	Standard Test Fluids Hydrocarbon and Iso-Octane
MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-G-5572	Gasoline, Aviation: Grades 80/87, 91/96, 100/130, 115/145

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MIL-J-5624	Jet Fuel, Grades JP-3, JP-4, and JP-5
MIL-R-6106	Relays, Electric, Aircraft, General Specification for
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-L-6880	Lubrication of Aircraft, General Specification for
MIL-F-7024	Fluids, Calibrating, for Aircraft Fuel System Components
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specification and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS20995	Wire, Lock
MS33586	Metals, Definition of Dissimilar
MS33588	Nuts and Plate Nuts, Self-Locking, Aircraft Design and Usage Limitations of

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PUBLICATIONS

Air Force - Navy Aeronautical Bulletin

No. 438 Age Controls for Synthetic Rubber Parts

Committee on Extension of Standard Atmosphere Publication

U. S. Extension to the ICAO Standard Atmosphere

(When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Attention : Code CDS.)

3. REQUIREMENTS

3.1 Preproduction. - The liquid level switch furnished under this specification shall be a product, samples of which have been tested and have passed the preproduction tests specified herein.

3.2 Selection of Government documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

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3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Metals - Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the switch may be subjected when in storage or during normal service life.

3.3.2.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in Standard MS33586 shall not be used in intimate contact with each other.

3.3.3 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the switch except where magnetic materials are essential.

3.3.4 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.5 Corrosive fumes - The materials used in the switch shall not liberate deleterious fumes.

3.3.6 All materials used in the sensing unit shall be sufficiently resistant to fluids conforming to Specifications MIL-J-5624, MIL-G-5572, and MIL-S-3136, Type I, of aromatic content from 0 to 30 percent, to assure satisfactory operation as herein defined.

3.4 Design and construction.

3.4.1 The basic system shall be designed to sense the presence or absence of liquids at definite levels and to actuate either the necessary warning or control circuits. It shall consist essentially of a sensing unit and a control unit mounted separately. The system shall be designed for reliable operation and rapid response. The sensing unit shall be designed so that the acceleration or sloshing of liquids, sound waves, light waves, or vibrations from external sources, or the adhesion of foreign materials will not cause the system to give a false indication of fuel level. Consideration shall be given to the necessity for sufficient pretravel and overtravel of moving parts and sufficient actuating force shall be available so that wear and other deterioration of such parts shall not induce failure during the normal life of the system. The design shall be reasonably simple to permit service maintenance, repair or overhaul without requiring special tools and fixtures. The sensing system shall be constructed to withstand the normal strains, jars, vibrations and such other conditions as are incident to shipping, storage, installation and service.

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3.4.2 The reduction of fire hazard shall be a major design consideration. Energy levels in the sensing unit shall be restricted to values that are intrinsically safe. The current in the circuits in or adjacent to a fuel cell shall not exceed 0.2 ampere for a maximum peak voltage of 165 volts for circuits of very low inductance. As a check on the inductance, the effective inductance in the circuit should be measured for the conditions most likely to give the maximum value in or adjacent to the fuel cell. The corresponding operating current (peak value for AC) should be determined and the energy value should be within one millijoule as determined by the formula:

$E = 1/2LI^2$ (E in millijoules)
L = inductance in millihenrys
I = peak current

For thermistor type systems the maximum surface temperature of the sensor for all specified test conditions shall be 316°C.

The above shall apply to abnormal conditions such as:

(a) Short circuiting of any capacitor or combinations of capacitors.

(b) Open circuit of any resistor or combinations of resistors or thermistors.

(c) All switches and other adjustments in the circuit set to any desired position.

3.4.3 Circuit protection - All thermistor type sensing units shall incorporate fuses or suitable protective devices to prevent excessive thermistor heating as a result of incorrect electrical installation or malfunction.

3.4.4 Component parts - The system shall consist of a minimum of two units. The sensing unit shall be the only component located in the fuel cell area. All other components shall be located in an area free from airfuel vapors.

3.4.5 Mounting - The design of the mounting shall depend upon the specific installation. Normal installation shall be either attachment to a tank unit probe or with an independent probe.

3.4.6 Operating voltage - The liquid level switch system shall operate satisfactorily over a range of voltages in accordance with Standard MIL-STD-704 and as specified by the procuring activity.

3.4.7 The sensor shall be designed so as to permit free entrance and drainage of liquid to and from the unit, and to prevent erratic operation due to the adhesion of liquid to the sensing element in any normal operating position.

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3.4.8 Buoyancy-float type - For float type switches, the float shall be designed to maintain adequate buoyancy in fluids with specific gravity of 0.6 or greater.

3.4.9 The control unit shall utilize a signal from the sensing unit to provide an output for the operation of the controlled device, such as indicators, valves and/or pumps. In systems having multiple sensing functions, the corresponding control units may be consolidated into a single or multiple package.

3.4.10 Chassis - The chassis and enclosure for component parts shall be of metal and shall be designed to form a rigid support for the equipment, protect it from changeable climatic and atmospheric conditions, and provide maximum accessibility for maintenance.

3.4.11 Vibration and shock insulation - For the protection of delicate components which may suffer damage or serious life reduction from the vibration or shock to be expected in service use, suitable cushioning devices shall be provided.

3.4.12 Reflection-ultrasonic type switches - Those switches which can be made to give a false indication of liquid level due to the reflection of ultrasonic waves shall be provided with an installation fixture to correct this condition.

3.4.13 Maintenance - The design of the system shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance using those tools and items of maintenance equipment which are normally available as commercial standards.

3.5 Performance - The system shall perform satisfactorily when subjected to the tests specified in Section 4.

3.6 Screw threads - Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.7 Electronic parts - Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on approved lists in accordance with Specification MIL-E-5400 shall not be used unless approved by the procuring activity.

3.7.1 Wire - Internal wire other than windings shall be in accordance with Specification MIL-W-16878.

3.7.2 Cable length - Accuracy and sensitivity of the equipment shall not be adversely affected by increasing or decreasing interconnection cable lengths within limits of 5 feet and 200 feet.

3.7.3 Connectors - The unit shall be designed for the minimum number of standard connectors. Connectors shall be polarized and shall be located for the minimum amount of attachment interference. The connectors shall conform to Specification MIL-C-5015.

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3.7.4 Grounding - Each electrical connector used in a fuel system component shall incorporate a ground return terminal.

3.7.5 Relays - All relays shall conform to Specification MIL-R-6106. They shall meet the requirements of the Sealing test and the Shock test of Specification MIL-R-6106 except that Procedure V of Specification MIL-E-5272 shall be used for the Shock test.

3.8 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.9 Physical requirements.-

3.9.1 Weight - The weight of the system, exclusive of wiring, shall be as low as practicable and shall not exceed 0.75 pound unless approved by the procuring activity.

3.9.2 Size - The size of the system shall be held to a minimum.

3.9.3 Finish - Protective coatings and finishes which will crack, chip or scale during normal service life or due to extremss of atmospheric conditions shall not be used.

3.9.3.1 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.9.3.2 Steel parts - Steel parts which are not corrosion resistant shall be cadmium plated, where practicable, in accordance with Specification QQ-P-416.

3.9.4 Soldering - Soldering shall be performed in accordance with Specification MIL-S-6872. No damage shall be caused to adjacent parts during the operating and there is to be no evidence of cold joints. Sufficient heat shall be applied so the solder flows freely and completely over the joint.

3.9.5 Lubrication - Lubrication shall be accomplished in accordance with Specification MIL-L-6880 except that the selection of the lubricant used shall be approved by the procuring activity.

3.9.6 Seal - In installations where the probe extends into a fuel cell from a flanged head external to the cell, the space within the head for wire connection shall be sealed to exclude fuel and moisture.

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3.9.7 Locking of parts - All threaded parts shall be positively locked by safety wiring, self-locking nuts, cotter pins, or other approved methods. Safety wire shall have a minimum diameter of 0.032 inch and shall conform to Standard MS20995. Self-locking nuts shall be used in accordance with Standard MS33588.

3.9.8 Riveting - Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.9.9 Dimensions - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practice. Where dimensions and tolerances may affect interchangeability, operation or performance of the instrument, they shall be held or limited accordingly.

3.10 Synthetic rubber parts -

3.10.1 Marking - All synthetic rubber parts such as diaphragms, but excepting "O" rings and parts with no suitable surface, shall have printed, stamped with ink, or otherwise noted on the part, the year and month of the curing date of the part.

3.10.2 Serviceability - All synthetic rubber parts shall be readily replaceable.

3.10.3 Uniformity - For components in contact with fuel, including parts fabricated of synthetic material, manufacturers shall control subsequent production to provide for uniformity.

3.10.4 Age controls - Age controls shall be in accordance with ANA Bulletin No. 438.

3.11 Identification of product - A suitable nameplate shall be securely attached to each separately mounted unit of the sensing system and shall be legibly and permanently marked in accordance with MIL-STD-130.

3.12 Installation -

3.12.1 Installation instructions - The contractor shall furnish with each system one copy of instructions, with illustrations and diagrams if necessary, covering the installation of the system. Prior to printing, two copies shall be furnished to the Bureau of Naval Weapons for approval. The instructions shall be printed on 8-1/2 by 11 inch or 11 by 17 inch paper.

3.12.2 Mounting hardware - The contractor shall furnish with each system sufficient mounting screws, bolts, nuts, etc. for system installation.

3.12.3 Envelope - An envelope furnished by the contractor containing installation instructions and mounting hardware shall be packaged with each system and shall be marked with the following information:

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IMPORTANT
THIS ENVELOPS CONTAINS INSTALLATION
INSTRUCTIONS AND MOUNTING HARDWARE

3.13 Workmanship - The switch and sensing unit, including all parts and accessories, shall be constructed and finished to produce a system free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests - The Inspection and Testing of the liquid level switch shall be classified as follows:

(a) Preproduction tests - Preproduction tests are those tests performed after award of contract on sample instruments that are representative in design, performance and configuration of the item which will be produced on the manufacturer's production line.

(b) Acceptance tests - Acceptance tests are those performed on instruments manufactured and submitted for acceptance under contract.

4.3 Preproduction tests - The preproduction tests of the liquid level switches shall consist of all of the tests of this specification specified under the paragraph headed Test Methods.

4.3.1 Preproduction Test Instructions - The preproduction test samples shall consist of three switches, of each manufacturer's part number to be furnished in accordance with the contract or order. The samples submitted shall not have been previously tested except for the Individual Tests. The manufacturer shall submit certified test results showing conformance with all the requirements and tests of this specification on samples other than the ones submitted for preproduction testing. The samples and certified test report shall be submitted to the testing activity designated in the contract or order.

4.3.2 Test Sample Identification - The test samples shall be plainly identified by securely attached, durable tags marked with the information listed below.

Sample for Preproduction Testing
Liquid Level Switch
Submitted by (Manufacturer's name, date) for Preproduction Testing in accordance with Specification MIL-S-21277 under contract or order number _____.

Manufacturer's Part Number

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4.4 Acceptance tests - The acceptance tests of the system shall consist of the individual tests and the sampling tests of this specification.

4.4.1 Individual tests - The individual tests of the system shall consist of the following tests conducted on each system:

Inspection
Leakage
Calibration-Room Temperature
Dielectric Strength

4.4.2 Sampling plans and tests -

4.4.2.1 Sampling plan A - Samples shall consist of three systems selected at random by the inspector from each lot of 100 or less systems which have passed the Individual tests. A lot shall consist of identical systems with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Systems which have been subjected to the Sampling plan A tests shall not be delivered on contract until they have been rebuilt and resubmitted to all individual tests.

4.4.2.1.1 Sampling plan A tests - The sampling plan A tests of the systems shall consist of the following tests conducted on each sample:

Low Temperature Operation
High Temperature Operation
Acceleration - Float Type Switches
Vibration
Voltage, Frequency and Temperature Variation

4.4.2.2 Sampling plan B - Unless otherwise specified, three systems selected at random by the inspector from the first 15 systems produced on contract and submitted within 10 days after manufacture shall be forwarded to the laboratory specified herein (see Section 6). Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

Submitted by (name) (date) for production test
in accordance with Contract No.

4.4.2.2.1 Sampling plan B test - The sampling plan B tests of the system shall consist of the following tests conducted on each sample:

Sampling Plan A
Altitude-Room Temperature
Fluid Dynamic Loading

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Surface Temperature - Thermistor Type
Sensing Element
Low Temperature Exposure
High Temperature Exposure
Thermal Shock
Fuel Shock
Non-Fuel Liquid Resistance
Fuel Resistance
Radio Noise Interference
Endurance
Humidity
Fungus Resistance
Salt Spray

4.5 Rejection and retest - The instructions on reworking or resubmitting rejected items shall be as specified in the general provisions of the contract.

4.6 Test conditions -

4.6.1 Atmospheric conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of Federal Standard No. 1.

4.6.2 Attitude - During the tests the sensor shall be oriented to positions which simulate possible operating positions in service.

4.6.3 Liquid adhesion-thermistor, optical and ultrasonic type switches - During tests which specify operation of the system, the system shall be operated with the sensing unit placed in the attitude or attitudes in which the test fluid would be most likely to adhere to the sensing unit so as to give false indications of liquid level.

4.6.4 Electrical power - Unless otherwise specified, the applied DC test voltage shall be 27.5 volts and the AC test voltage shall be 115 volts, 400 cycle, single phase. The voltage tolerance shall be ±2 percent.

4.6.5 Test fluid - Unless otherwise specified herein, the fluid used for testing systems which are designed for use in hydrocarbons shall be fuel conforming to Specification MIL-G-5572, fuel conforming to Specification MIL-J-5624, Grades JP-4 and JP-5, test fluid conforming to Specification MIL-S-3136, Type I, or other fluids which conform to Specification MIL-F-7024. The fluid to be used for testing those systems which are designed for use in liquids other than hydrocarbons, shall be specified by the procuring activity.

4.6.6 Altitude - Altitude pressures shall be determined from U. S. Extension to the ICAO, Standard Atmosphere, Tables and Data, for applicable altitudes.

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4.6.7 Cleaning - Prior to testing any component part, all lubricants and preservative compounds, except permanent protective coatings, such as paint, shall be removed from the external parts of the unit and from all internal parts which are normally wetted with fuel.

4.7 Test methods -

4.7.1 Inspection - Each system component shall be examined externally to determine conformance to the applicable drawings and to all the requirements of this specification not covered by tests.

4.7.2 Leakage - Sensing units of the externally mounted type shall be installed in a suitable pressure chamber equipped with the proper size flange fittings. A pressure of 30 ± 5 psi shall be applied within the pressure chamber for 5 minutes or as designated by the procuring activity. The unit shall be tested for leakage by applying a suitable liquid to the exposed part. The development of bubbles shall be cause for rejection.

4.7.3 Calibration-room temperature -

4.7.3.1 Thermistor type switches - With the sensing element in air, the system shall be electrically energized and allowed to stabilize for one minute. The sensing unit shall then be lowered into a container of test fluid until the sensing element is approximately 1/4 inch above the liquid surface. The sensing element shall then be immersed quickly into the fluid and the time required for the system to operate shall be recorded. After being immersed for 10 seconds, the sensing unit shall be withdrawn quickly and the time required for the reverse operation shall be recorded. This procedure shall be repeated at least five times and the average operating time (herinafter called response time) shall not exceed the requirements of Table I.

Table I. Response time tolerances for thermistor type switches

Input	Room Temperature (25° ± 2°C)		Low Temperature (-55° ± 2°C)		High Temperature (70° ± 2°C)	
	Air to Liquid	Liquid to Air	Air to Liquid	Liquid to Air	Air to Liquid	Liquid to Air
20V DC	1 sec	30 sec	3 sec	45 sec	1 sec	30 sec
22V DC	1 sec	20 sec	3 sec	30 sec	1 sec	20 sec
24V DC	1 sec	10 sec	3 sec	15 sec	1 sec	10 sec
26V DC	1 sec	10 sec	3 sec	15 sec	1 sec	10 sec
28 V DC	1 sec	10 sec	3 sec	15 sec	1 sec	10 sec
29 V DC	1 sec	10 sec	3 sec	15 sec	1 sec	10 sec

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4.7.3.2 Float type switches - These switches shall be tested as in 4.7.3.1 except that the 10-second stabilization period may be omitted and the response time shall not exceed one second when the sensing element is immersed in, or withdrawn from, the test fluid.

4.7.3.3 Ultrasonic and optical type switches - These switches shall be tested as in 4.7.3.1 except that the response time shall not exceed one second when the sensing element is immersed in or withdrawn from the test fluid.

4.7.4 Dielectric strength - The applicable voltage at commercial frequency as shown on Table II shall be applied between circuits and between each circuit and the metal frame of the component for a minimum period of one minute. Current flow in excess of 2 milliamperes or breakdown of insulation shall constitute a failure. The test voltage shall not be applied to test points between which there are electrical circuit elements that would be damaged by the test voltage.

TABLE II

Nominal Voltage	Test Voltage
27.5V DC	500V
115V 400-cycle AC	750V

Where hermetically sealed enclosures are employed, the above test shall be conducted prior to sealing the enclosures. After the enclosures are sealed and pressurized, the same test shall be repeated with the exception that the test voltage shall be reduced to 200 volts.

4.7.5 Low temperature operation - The system shall be placed in a temperature chamber and subjected to a temperature of -55 ± 2°C for a minimum period of 4 hours. At the end of this period and while the units are still at the low temperature the Calibration test of 4.7.3 shall be performed. The response time for thermistor type switches shall not exceed the requirements of Table I. Float type, ultrasonic type, and optical type switches shall operate within one second after being immersed in or withdrawn from the test fluid.

4.7.5.1 When the units have returned to room temperature, the Calibration test of 4.7.3 shall be repeated.

4.7.6 High temperature operation - The system shall be placed in a temperature chamber and subjected to a temperature of 70 ± 2°C for a minimum period of 4 hours. At the end of this period and while the units are still at the elevated temperature the Calibration test of 4.7.3 shall be performed. The response time shall not exceed the requirements of Table I for thermistor type switches and one second for float type, ultrasonic type, and optical type switches.

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4.7.7 Acceleration-float type switches - The system shall be mounted on a centrifuge along with a battery and indicator light wired so that movement of the float will be indicated. The mounting position of the switch shall be such that the plane through which the float moves is parallel to the plane of rotation of the centrifuge and so that the accelerating force is toward the top side of the switch. An accelerating force of 10 g's shall be attained, stabilized and maintained for one minute. There shall be no change in light indication from 0 to 10 g.

4.7.7.1 The position of the switch shall be reversed so that the applied force is toward the bottom side of the switch and the procedure of 4.7.7 shall be repeated. There shall be no change in the light indication during this test.

4.7.7.2 Following the tests of 4.7.7 and 4.7.7.1 the Calibration test of 4.7.3 shall be performed. There shall be no malfunction or evidence of mechanical failure.

4.7.8 Vibration - The system shall be subjected to vibration in accordance with Procedure XII of Specification MIL-E-5272. The control units shall be vibrated with actual operating loads connected to them. The system shall operate satisfactorily during and after the Vibration test and shall show no evidence of mechanical failure.

4.7.9 Voltage, frequency and temperature variation - The system shall operate satisfactorily under all combinations of the voltage, frequency, and temperature extremes of Table III. The tolerances of 4.7.3 shall apply.

Table III

Voltage	Frequency	Temperature
(AC Systems) 105 to 125V	360 to 440 cps	-55 ± 2°C to 70 ± 2°C Air -55 ± 2°C to 70 ± 2°C Liquid
(DC Systems) 20 to 29V		-55 ± 2°C to 70 ± 2°C Air -55 ± 2°C to 70 ± 2°C Liquid

4.7.10 Altitude-room temperature - The system shall be maintained at a temperature of 25 ± 2°C and a pressure of 0.82 inch Hg absolute 80,000 feet for 15 minutes, during which time the sensing element is immersed in test fluid. While at this temperature and simulated altitude the applicable test of 4.7.3 shall be performed and the response time limits of that paragraph shall apply. There shall be no malfunction or evidence of mechanical failure.

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4.7.11 Fluid dynamic loading - The portion of the sensing unit projecting into the fuel tank shall be tested as follows: A load of 5 pounds per square inch of projected area shall be applied alternately on opposite sides of the unit normal to the major axis at a rate of 12 cycles per minute for a 10-hour period. At the end of this period there shall be no sign of mechanical failure. The Leakage test of 4.7.2 and the Calibration test of 4.7.3 shall be repeated.

4.7.12 Surface temperature-thermistor type sensing element - The sensing element shall be immersed in a bath of silicone oil or other suitable liquid and heated externally. At every 10-degree interval from 70° to 230°C, a small current shall be passed through the sensing element to simulate a self-heating effect and its resistance shall be measured. These values shall be plotted and a resistance-temperature curve obtained. The sensing element shall then be connected to the other components of the system and the system shall be energized. While the system is in operation in an air temperature of 70°C, the resistance of the sensing element shall be measured. This value shall be applied to the resistance-temperature curve and the operating temperature determined. This temperature shall not exceed 316°C.

4.7.13 Low temperature exposure - The assembly shall be tested as specified in 4.7.5 except that the assembly shall be maintained at a temperature of -65 ± 2°C for a period of 48 hours, and then the temperature shall be elevated to -55 ± 2°C for a 24-hour period. During the last 4 hours at -55°C, the pressure shall be reduced to 0.82 inch Hg absolute, or less. At the completion of the 24-hour period and while at the temperature and pressure specified above, the Calibration test of 4.7.3 shall be repeated and the response time limits of that paragraph shall apply.

4.7.13.1 When the units have returned to room temperature the Calibration test of 4.7.3 shall be repeated.

4.7.14 High temperature exposure - The assembly shall be tested as specified in 4.7.6 except that the assembly shall be maintained at a temperature of 70 ± 2°C for a period of 24 hours. At the end of this period and while still at the elevated temperature the Calibration test of 4.7.3 shall be performed and the response time limits of that paragraph shall apply.

4.7.14.1 When the units have returned to room temperature the Calibration test of 4.7.3 shall be repeated.

4.7.15 Thermal shock - The assembly shall be placed in a test chamber and maintained at a temperature of -55°C for one hour. At the end of this time it shall be removed from this test chamber and placed in a test chamber with a temperature of 70°C and left for one hour. The time involved in changing from one environment to the other shall not exceed 2 minutes. The cycling shall be conducted four times. During this test the assembly shall be energized at rated operating voltage. There shall be no evidence of malfunction or physical damage as a result of temperature changes. The assembly shall then be allowed to return to room temperature (25°C) and the Calibration test of 4.7.3 shall be repeated.

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4.7.16 Fuel shock -

4.7.16.1 Cold air to hot fuel - The assembly shall be placed in a test chamber and maintained at a temperature of -55°C for one hour. At the end of this period it shall be removed from the test chamber and energized at rated operating voltage. The sensing unit shall be immersed in test fluid maintained at 70°C until the circuit operates, after which it shall be removed from the test fluid. One reading for both "air to liquid" and for "liquid to air" response times shall be recorded, the assembly de-energized and returned to low temperature. The cycling shall be conducted a total of eight times. During the test there shall be no evidence of malfunction.

4.7.16.2 Hot air to cold fuel - The assembly shall be placed in a test chamber, energized at rated operating voltage and maintained at a temperature of 70°C for one hour. At the end of this period it shall be removed from the hot air, and the sensing unit shall be immersed in test fluid maintained at -55°C until the circuit operates, after which it shall be removed from the test fluid. One reading for both "air to liquid" and "liquid to air" response times shall be recorded and the assembly returned to high temperature. This cycling shall be conducted a total of eight times. During the test there shall be no evidence of malfunction.

4.7.17 Non-fuel liquid resistance - Systems designed for use with liquids other than fuel shall be operated with the sensing unit in the fluid in which it is to be used, at a temperature and for a period of time to be determined by the qualifying activity. Following this the Calibration test of 4.7.3 shall be performed.

4.7.18 Fuel resistance - The sensing unit shall be soaked for 68 hours in Type II fuel of Specification MIL-S-3136 at a temperature of 70 ± 2°C. The unit shall then be removed from the fuel and dried at a temperature of 70 ± 2°C for a period of 4 hours. This cycling shall be conducted a total of three times. After the final drying out period, the Calibration test of 4.7.3 shall be repeated.

NOTE: During the period of soaking in test fluid, all non-magnetic parts that would be exposed to fuel in service use shall be maintained in such a manner as to insure complete contact with the fluid.

4.7.19 Radio noise interference - The radio noise suppression test shall be in accordance with Specification MIL-I-6181.

4.7.20 Endurance - The endurance test shall be conducted by cycling the sensing units between liquid and air according to the following schedule, unless otherwise specified by the procuring activity:

90 percent of the cycles with room temperature air and liquid
5 percent of the cycles with liquid and air at -55 ± 2°C
5 percent of the cycles with liquid and air at 55 ± 2°C

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The cycling may be accomplished automatically by means of the level system using all of the components of the system. Relay contacts and solid-state relays shall be operated at maximum load condition for 100,000 complete cycles. The endurance test shall be conducted using any fluid acceptable to the procuring activity, preferably that fluid normally used with the component in service use. The cycling rate shall be adjusted so that the duration of each cycle is approximately 5 seconds more than the response time specified for the particular temperature at rated voltage. Following the endurance test the Calibration test of 4.7.3 shall be repeated.

4.7.21 Humidity - The humidity test shall be conducted in accordance with Procedure I of Specification MIL-E-5272 except that the test period shall be 240 hours (10 cycles). Within one hour after exposure to humidity, the Calibration test of 4.7.3 shall be repeated. The equipment shall also be visually inspected for any deterioration or corrosion which could in any manner prevent the equipment from meeting the operational, maintenance, and service requirements specified herein.

4.7.22 Fungus resistance - The fungus resistance test shall be conducted in accordance with the requirements of Specification MIL-E-5272 except that the duration of the test shall be 14 days. At the conclusion of the test the equipment shall be visually examined to ascertain that no fungus growth has occurred.

4.7.23 Salt spray - The system shall be placed in its normal operating condition in a slat spray chamber and subjected to a salt spray in accordance with Procedure I of Specification MIL-E-5272 for a period of 50 hours. Parts of the system which would not normally be subjected to environmental conditions in service use shall be covered. At the end of the 50-hour period the equipment shall be examined. There shall be no evidence of corrosion or deterioration which will affect subsequent operation of the equipment.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The systems shall be packaged in accordance with Specification MIL-P-7936, Level A or B as specified in contract or order (see 6.2). For Level A packaging the method of preservation shall be in accordance with MIL-P-116 Method 1A-5 without preservation compound, using metal reusable containers.

5.2 Packing - The systems shall be packed in accordance with Specification MIL-P-7936, Level A, B or C as specified in contract or order (see 6.2).

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

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6. NOTES

6.1 Intended use - The liquid level switches covered by this specification are intended for use in sensing liquid level and providing remote indication or control thereof.

6.2 Ordering data - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether Sampling Plan B is to be omitted, or name and address of the laboratory where amsples shall be submitted.

c. Levels of packaging and packing desired.

6.3 Drawing Requirements.- The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

6.4 Provisions for Preproduction Tests.- The manufacture of switches on contract shall not commence until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of the switches from contract to contract, submission of further preproduction samples on the subsequent contracts may be waived at the descretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirement of acceptance meeting.

6.4.1 It shall be understood that the switch supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the procuring activity.

NOTICE - When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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